

MATTHEW L BROWN

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SUMMARY

Full-stack developer with 10+ years of experience designing data-intensive applications, analytics platforms, and cloud infrastructure. I follow a customer-focused approach to deliver solutions that produce measurable results.

FREELANCE

Software Consultant / Solutions Architect

July 2019 - Present

Selected Client Projects

Population Health Analytics Tool [R, SQL, JavaScript, Google Cloud]

Reduced analyst time spent on recurring ad-hoc data requests by enabling self-service analysis

- Led development of a population health analytics platform used by thousands of users at the org
- Developed a data pipeline to process terabytes of data from 10+ sources into **BigQuery** analytics layer
- Enabled researchers and analysts to perform cohort analysis and real-world evidence studies.

Field Operations Analytics Platform [dbt, Python, React, Azure]

Enabled the client to shift from hourly to fixed-price projects by accurately modeling field productivity and costs.

- Built a geospatial data pipeline using dbt and Python to turn raw GPS telemetry into analytics tables
- Developed a classification model to measure field productivity from GPS and GIS data
- Designed a React app for reviewing work progress and cost forecasts

Reinsurance Claims Analytics Tool [Python, SQL, MongoDB, React, TypeScript, Google Cloud]

Saved hours each week from the manual process of ingesting new client data

- Designed and developed an analytics platform for insurance claims, weather, and geospatial data.
- Built Python data pipelines to ingest claims data, weather data, and shapefiles into a SQL-based platform.
- Used GeoPandas and Shapely to integrate GIS datasets for catastrophe risk analysis.
- Developed flexible dashboards and optimized SQL queries for real-time access to large datasets

Digitizing Iowa Plastics Manufacturer [PowerApps, R, Fabric]

Reduced scrap by 10% and excess material usage by 60% through improved tracking and visibility.

- Built a QR-based traceability system linking production, quality, inventory, and shipping data.
- Developed real-time inventory tracking and automated label printing for shop-floor operations.
- Created operational dashboards for production, downtime, quality, and inventory performance.

Multi-Store E-Commerce Analytics Platform [Snowflake, dbt, Airbyte]

Built an end-to-end data pipeline consolidating data from multiple e-commerce shops into a unified analytics layer

- Integrated Shopify, GA4, Meta Ads, and Google Ads data with Airbyte.
- Built and tested dbt models for portfolio-level revenue, customer, product, and marketing analytics.
- Developed Lightdash dashboards for portfolio and individual store performance.

Healthcare Platform Cloud Migration [Terraform, GCP, Azure, Python]

Identified and remediated critical security risks, including publicly exposed SQL instances.

- Led migration of multi-service healthcare platform (App Engine, GKE, Cloud Run, Cloud SQL, Pub/Sub) to Azure
- Re-architected networking with Azure Front Door, VNet integration, private endpoints, and Managed Identity.
- Implemented GitHub Actions CI/CD and modular Terraform IaC, enabling repeatable deployments.

Business Rule Engine for Clinical Interventions [R, SQL, JavaScript]

Client commercialized the product and offered as a service to other healthcare clients.

- Developed a database-driven system to identify members for clinical interventions
- Designed an intuitive GUI interface so non-technical users could define member cohorts without code.
- Built internal tooling that significantly reduced development time for new intervention cohorts.
- Built the core logic supporting a ~50-person team; the architecture remained in production without significant modification for 6+ years.

CRM Application Suite Optimization for a Health Insurer [R, SQL, JavaScript]

Improved application load times by 10x through optimizing SQL queries.

- Centralized duplicate code across the applications into a repository.
- Re-architected data model for simplicity and maintainability

FOUNDER

Notifi: Back in Stock Alerts

Nov. 2020 - Oct. 2025

*Built and operated a Shopify SaaS application for back-in-stock notifications with **1000+ installs**, **600+ active users**, and **\$1000+ MRR***

Echo - Speech-to-Text Engine

June 2024 - Dec 2024

Built an AI-powered Chrome extension for real-time speech-to-text transcription.

Avisos: PPC Management Copilot

Dec. 2025 - May 2026

Avisos connects your Google Ads account and Shopify store to audit 50+ performance, targeting and tracking issues.

EMPLOYMENT

RED OAK STRATEGIC

Senior Data Scientist

Washington, D.C.

April 2017- June 2019

Developed analytics software for financial services startup

- Led a team of three in developing 30+ custom analytical applications using **R Shiny**
- Implemented the best analytical dashboard into the production **.NET** application
- Developed predictive models on large datasets using machine learning techniques

BOOZ ALLEN HAMILTON

Senior Consultant

Washington, D.C.

June 2015- April 2017

Management consultant focused on implementing large-scale technology projects for Federal clients

- Developed strategies to maximize adoption of **agile software development** best practices as a **project manager**
- Scrum master for an **agile software development** project to migrate [DOL.gov](https://www.dol.gov/) to Drupal

JOHN DEERE

Geometric Coordinator

Waterloo, IA

Nov. 2011-Aug. 2013

Quality analyst for design engineering division

TECHNICAL SKILLS

Languages: Python, R, SQL, TypeScript, JavaScript, C#

Frameworks: Django, React, FastAPI, R Shiny, .NET, React

Databases: PostgreSQL, BigQuery, SQL Server, MongoDB, dbt

Cloud & DevOps: Google Cloud, Azure, AWS, Terraform, Docker, GitHub Actions

Coursework / Certifications: *Introduction to R for Data Science, The Analytics Edge, Explore Data Science, Advanced Analytics, Statistical Methods of Process Improvement*

EDUCATION

HENRY B. TIPPIE SCHOOL OF MANAGEMENT, UNIVERSITY OF IOWA
Master of Business Administration, Supply Chain and Analytics

Iowa City, IA
May 2015

UNIVERSITY OF IOWA
BSE Biomedical Engineering

Iowa City, IA
May 2011